

COURSE TOPICS

- **Disinformation.**
- **Data and consent.**
- **Algorithmic bias and fairness.**
- **Women and gender.**
- **Race.**
- **Social network data.**
- **Privacy, surveillance, transparency and safety.**
- **Ethics and governance of AI and big data.**
- **Issues from data mining, data exploration and data collection.**
- **Algorithmic decision making.**
- **AI and health care.**
- **Malicious uses of AI.**
- **Racist AI.**
- **AI algorithm (NLP, Computer vision)**
- **AI for social good.**
- **AI and international security.**
- **Autonomous weapons**
- **Teaching ethics to robots**
- **Robot rights.**
- **The Internet of Things, Smart Homes, and Smart Cities.**



GOALS

- By the end of the course, the student should be able to:
 - Demonstrate knowledge of philosophical issues involved in ethics of data science and cybersecurity.
 - Demonstrate familiarity with relevant examples of AI systems
 - Show ability to work in a small team
 - Show ability to produce written work regularly to a deadline
 - Acquire ability to express arguments clearly and concisely
 - Gain skills in research, analysis and argumentation



Topic1

ETHICAL & SOCIAL ISSUES IN THE INFORMATION AGE: AN INTRODUCTION



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Chapter 1, 2 & 3 in “Ethical & Social Issues in the Information Age”.

Chapter 1& 2 in “Computer Ethics: A global perspective”.

INFORMATION TECHNOLOGY SOCIAL SIGNIFICANCE

- The rapid advance in information Technology (IT) has brought modern society into the “***information age***”.
- *This revolution **surpasses** both the **agricultural** and **industrial** revolutions in **social significance**.*
- **Why?**
- *A new reality called “**Information Economy**”.*



INFORMATION TECHNOLOGY SOCIAL SIGNIFICANCE (CONT.)

- ***Information economy*** enables people to:
 - Earn their living by **working digitally** with money and ideas.
 - **Exchange knowledge & information** in **digital** form rather than **physical** goods, tools, and services.
- The increased role of information activities & industry rapidly affects **the way people live, communicate, work, and entertain.**
- Humans are **widely connected** to and sometimes totally **dependent** on information systems & computing tools.



INFORMATION TECHNOLOGY ETHICAL ISSUES

- IT has become a principle tool for political power, authority, and propaganda, **as well as** for providing the **opportunity for electronic crimes, fraud, surveillance & security.**
- In this multi-faceted world of information society, **traditional ethical human behavior evolve.**
- Traditional ethical questions are **re-examined** and new questions arise concerning **moral standards** for human behavior.



INFORMATION TECHNOLOGY ETHICAL ISSUES

- Information systems raise new ethical questions because they create opportunities for:
 - **Intense social change**, threatening existing distributions of power, money, rights, and obligations.
 - New kinds of crimes.



PART-I OF THE THIS COURSE: ETHICAL ISSUES IN COMPUTING.

- We provide an introduction to the **ethical and social issues** sparked by our ever-growing **information society**.
- We consider the **impact** of digitized information on **individuals** and **societies** and include discussions on **privacy, reliability, security, intellectual property**.
- We also discuss the **social impact of information technology** in different areas of human life.



TERMINOLOGIES

- **Information Technology (IT)**
 - Information Technology is the application of **appropriate technologies** to the **organization, manipulations & distribution of information** by **computers & telecommunications**.



ETHICS

○ **Ethics**

- From a Greek word “ethos”: character/customs.
- A branch of philosophy.
- Involves:
 - Freedom, privacy, equality, duty, obligations choice, justification of judgments, rights, claims



MORALITY

- A set of rules of right conduct
- A system used to modify and regulate our behavior
- The code or set of principles, standards, or rules that guide the moral actions of an individual within a particular social framework.



DEFINITION OF MORALITY

- Moral judgment & decision: how a person ought to act & the extent to which the action is described as right or wrong.
- Morality includes virtues like:
 - Love for others
 - Compassion.
 - Desire for justice.

MORALITY

- It builds character traits in individuals.
- It is **group-based** in the sense that it is a set of shared rules, principles and duties applicable to a group/society and independent of religion.
- It has no reference to social standing of **individuals** in the **group**
- It is **influenced** by other factors like:
 - Time
 - place

MORALITY AND ETHICS

- Ethics is generally defined as: '**Systematic study of morality**'
- Ethics is a **study** of right & wrong in human conduct.
- A theoretical **examination of morality** & as an equivalent of the theory of moral.



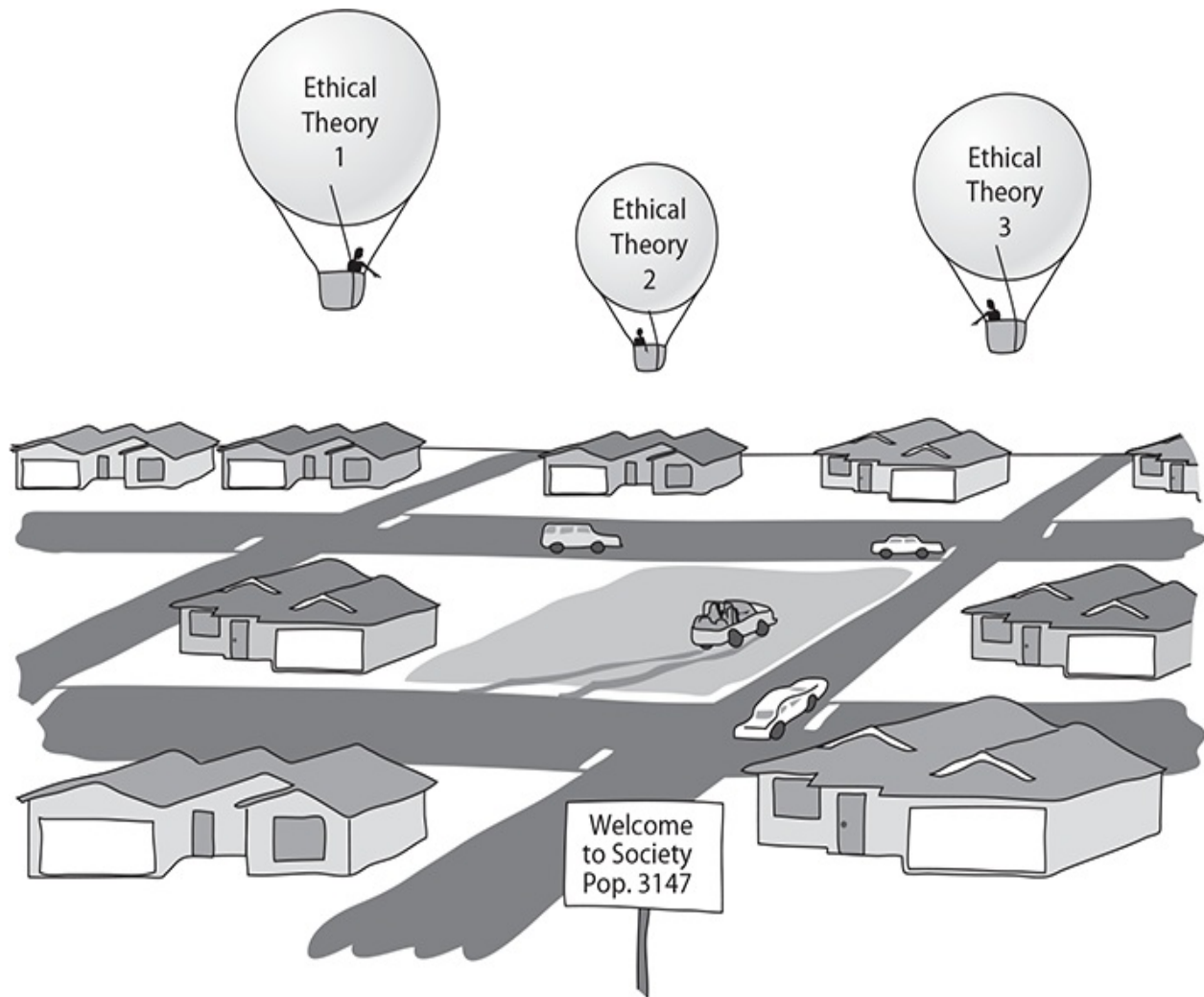


Figure 2.2

An analogy explaining the difference between ethics and morality. Imagine society as a town. Morality is the road network within the town. People doing ethics are in balloons floating above the town.

systems). While there may in fact be a definite answer regarding the best way to construct and operate a road network, it may be difficult for the observers to identify and agree upon this answer, because each observer has a different viewpoint.

MORALITY VS. ETHICS: COMPARISON CHART

	Ethics	Morals
What are they?	The rules of conduct recognized in respect to a particular class of human actions or a particular group or culture.	Principles or habits with respect to right or wrong conduct. While morals also prescribe dos and don'ts, morality is ultimately a personal compass of right and wrong.
Where do they come from?	Social system – External	Individual - Internal
Why we do it?	Because society says it is the right thing to do.	Because we believe in something being right or wrong.
Origin	Greek word "ethos" meaning "character"	Latin word "mos" meaning "custom"
Acceptability	Ethics are governed by professional and legal guidelines within a particular time and place	Morality transcends cultural norms



COMPUTER ETHICS

- The **systematic study** of the **ethical & social** impact of computer in the **information society**.
- Involve the acquisition, distribution, storage, processing & dissemination of digital data & how individuals & groups interact with the systems & data.
- Responsibility & accountability.
- Policies, regulations, legislation.



COMPUTER ETHICS: MOORE'S DEFINITION

- **James H. Moore** is believed to have **first** coined the phrase "computer ethics".
- According to Moore, computer ethics is the analysis of the nature and social impact of computer technology and the corresponding formulation and justification of policies for the ethical use of such technology .
- Moore's definition **focuses** on the human actions that are routed in computer technology or influenced by computer technology.



COMPUTER ETHICS...

- In other words, it is a study, an analysis of the values of human actions influenced by computer technology.
- Computer influence on human actions is widespread throughout the decision making process preceding the action.
- We are looking for a way to deal with these problems, probably through education.
- So the definition of computer ethics, as outlines by Moore, gives us a starting point to our study.



COMPUTER CRIMES

- The illegal activities related to computers and computer networks.
- Any illegal act for which knowledge of computer technology is used to commit the offence. [National Institute of Justice]
- Any criminal act that has been committed using a computer as the principle tool.
- Some of the computer crimes are **transformation of old crimes** others are **entirely new kinds of crimes**.



THE EMERGENCE OF THE SOCIAL AND ETHICAL PROBLEMS IN COMPUTING

- One can perhaps say that the history of computer crimes started with the invention of the **computer virus**.
- A **computer virus** is defined as:
 - A **self-propagating** computer program designed to **alter** or **destroy** a computer system resource,
 - It follows almost the same pattern but instead of using the living body, it uses software to attach itself, grow, reproduce and spread in the new environment.



COMPUTER VIRUS

- As a virus spreads in the new environment, it attacks major system resources that include the surrogate **software** itself, **data**, and sometimes **hardware** resources
- **Aims to damage** the capacity of these resources to perform the needed functions and eventually bringing the system down.
- The word virus was first assigned a non-biological meaning in the 1972 science fiction stories about the G.O.D.



WHY THE NEED FOR NEW STUDY?

- New Technology provides New Choices & Abilities
 - Behavior
 - Public Policies
 - Personal Policies



IS COMPUTER ETHICS UNIQUE?

- Considering the following two schools of thought. Which one is right? Why?
- School-1:
 - believes in the study of computer ethics as remedial moral education.
- School-2:
 - believes in computer ethics education not as a moral education but as a field worthy of study in its own right as a separate independent field of study,
 - there must be a unique domain for computer ethics distinct from the domain for moral education, distinct even from the domains of other kinds of professional and applied ethics.



IS COMPUTER ETHICS UNIQUE?

- In his paper “**Is Computer Ethics Unique?**”, Walter Maner explained:
 - Certain ethical issues are so **transformed** by the use of computers that they **deserve** to be studied on their own, in their **radically altered form**,
 - The involvement of computers in human conduct can create **entirely new ethical issues, unique** to computing, that do not surface in other areas.



WALTER MANER'S JUSTIFICATION

1. We should study computer ethics because doing so will make us **behave** like responsible professionals.
2. We should study computer ethics because doing so will teach us how to **avoid** computer abuse and catastrophes.
3. We should study computer ethics because the advance of computing technology will **continue** to create **temporary policy gaps**.



WALTER MANER'S JUSTIFICATION

4. We should study computer ethics because the use of computing **permanently transforms** certain ethical issues to the degree that their alterations **require independent study**.
 5. We should study computer ethics because the use of computing technology **creates**, and will **continue** to create, novel ethical issues that require special study.
 6. We should study computer ethics because the set of **novel** and **transformed** issues is **large enough** and coherent enough to define a new field.
- **Whatever school-1 falls in, there is enough justification to study computer ethics.**



DISCUSSION QUESTIONS

- IS your chosen crime a transformation of old crimes or a unique (entirely new) computer crime?
- Compare Morality to Ethics.
- 3 lessons (comments) from TED's Talk:
<https://www.youtube.com/watch?v=cf3zxHuSM2Y>

